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Rocket injector subscale stability assessment with telescoping piston

Abstract

Systems and methods for subscale testing of rocket engine injector stability. The system includes a combustion chamber and a piston within the chamber that is continuously axially moveable via an actuator. An annular gap between the piston and a chamber sidewall provides a minimal cross-sectional flow area. A modular injector plate comprises one or more injector elements configured to inject a fuel and an oxidizer into the chamber. The piston is continuously translated, to thereby continuously vary a combustion volume of the chamber and to create a dynamically tunable downstream boundary. The injector elements are thus exposed to acoustic modes of varying frequency, covering the range of acoustic modes expected in a full scale rocket engine. The injector plate is removably attached to an upstream end of the chamber for replacement of the first injector elements with different, second injector elements for subsequent testing.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. For example, this application claims the benefit of U.S. Provisional Application No. 63/477,633, filed on Dec. 29, 2022, titled SUBSCALE STABILITY ASSESSMENT OF ROCKET ENGINE INJECTORS, the entire contents of which is incorporated by reference herein and forms a part of this specification for all purposes.

BACKGROUND

Field

(1) The development relates generally to rocket engines, in particular to subscale testing of rocket engine injector stability.

Description of the Related Art

(2) Combustion stability of rocket engine injectors is critical for engine performance. However, combustion stability is poorly understood and lacks predictive analytic models. Further, stability is a system level phenomenon and is thus a function of full-scale engine combustion chamber modes. Stability issues therefore often only become apparent during full-scale tests, where iteration on designs is costly and time-consuming, along with placing risk on other engine components. Conventional subscale testing capabilities of rocket injectors are limited and do not produce sufficient stability assessment of full-scale performance. Improvements to these and other drawbacks of existing solutions for subscale testing of injector stability are therefore desirable.

SUMMARY

(3) The embodiments disclosed herein each have several aspects no single one of which is solely responsible for the disclosure's desirable attributes. Without limiting the scope of this disclosure, its more prominent features will now be briefly discussed. After considering this discussion, and particularly after reading the section entitled "Detailed Description," one will understand how the features of the embodiments described herein provide advantages over existing systems, devices and methods for subscale testing of injector stability.

(4) The following disclosure describes non-limiting examples of some embodiments. For instance, other embodiments of the disclosed systems and methods may or may not include the features described herein. Moreover, disclosed advantages and benefits may apply only to certain embodiments of the invention and should not be used to limit the disclosure.

(5) Systems, devices, and methods are described for subscale testing of rocket engine injector stability. The system includes a combustion chamber and a piston within the chamber that is

continuously axially moveable by an actuator. An annular gap between the piston and chamber sidewall provides a minimal cross-sectional flow area. A cross-sectional area of the flow path in the downstream direction may decrease to a minimum at the annular gap about the piston and then increase. A modular injector plate comprises one or more first rocket engine injector elements configured to inject one or more propellants into the chamber. The piston is continuously translated to thereby continuously vary a combustion volume of the chamber and to create a dynamically tunable downstream boundary. The injector elements are thus exposed to acoustic modes of varying frequency, covering the range of acoustic modes expected in a full scale rocket engine. The injector plate is removably attached to an upstream end of the chamber for replacement of the first injector elements with different, second injector elements for subsequent testing. Sensors in the chamber detect the acoustic responses. A cavity ring forming part of the chamber may be replaceable for testing various components and parameters, such as film cooling, alternative cavity ring designs, etc. Particular aspects and embodiments thereof are summarized below, followed by detailed discussion of these and other features.

(6) In one aspect, a system for subscale testing of rocket injector stability is described. The system comprises a chamber, an injector plate, and a piston. The chamber has a sidewall extending axially. The injector plate is removably attached to an upstream end of the chamber. The injector plate comprises one or more injector elements configured to inject one or more propellants into the chamber. The injector plate is one of a plurality of different injector plates configured to be removably attached to the chamber. The piston is positioned at least partially within the chamber. The piston is continuously moveable in an axial direction to continuously vary a combustion volume of the chamber located between the injector plate and an upstream end of the piston.

(7) Various embodiments of the various aspects may be implemented. In some embodiments, the upstream end of the piston is rounded. In some embodiments, the system further comprises an actuator configured to axially move the piston. In some embodiments, the piston further comprises a cooling channel. In some embodiments, the system further comprises a shell at the upstream end of the piston, wherein the cooling channel extends between the shell and the piston. In some embodiments, the system further includes a shell at the upstream end of the piston. The shell and the piston may be monolithic. In some embodiments, the piston and the sidewall of the chamber define a gap therebetween. In some embodiments, the gap is an annular gap. In some embodiments, the piston increases in cross-sectional area in the downstream direction from an upstream end of the piston to a maximum cross-sectional area at the annular gap. In some embodiments, the piston decreases in cross-sectional area in the downstream direction from the annular gap. In some embodiments, the system further comprises a sensor configured to detect a pressure within the combustion volume. In some embodiments, the system further comprises a replaceable cavity ring forming part of the sidewall of the chamber.

(8) In another aspect, a subscale rocket injector stability test system includes a chamber, a modular injector plate, and a piston. The modular injector plate is coupled to the chamber and comprises one or more injector elements. The piston is positioned at least partially within the chamber and is configured to move to vary a combustion volume of the chamber. The piston and a sidewall of the chamber define an annular flow path around the piston configured to act as a chokepoint allowing exhaust to exit the chamber.

(9) Various embodiments of the various aspects may be implemented. In some embodiments, an end of the piston facing the combustion volume is bulbous. In some embodiments, the piston further comprises a cooling channel extending at least partially within an extension rod coupled to the piston and at least partially within the piston. In some embodiments, the chamber is cylindrical. In some embodiments, a volume defined between the piston and the sidewall defines an inverted nozzle shape. In some embodiments, the system further comprises an actuator configured to axially move the piston to continuously vary the combustion volume.

(10) In another aspect, a method of subscale testing rocket injector stability includes injecting into a

combustion chamber one or more propellants from one or more injector elements supported by a modular injector plate to cause combustion, continuously moving a piston within the combustion chamber to continuously vary a combustion volume within the chamber, and detecting data from within the combustion volume as the piston moves.

(11) Various embodiments of the various aspects may be implemented. In some embodiments, the method further includes flowing a coolant through a cooling channel within the piston. In some embodiments, the method further includes replacing the injector plate with a different injector plate supporting one or more different injector elements. In some embodiments, the method further includes flowing exhaust from the combustion through an annular gap located between the piston and a sidewall of the chamber. In some embodiments, the method further includes analyzing the acoustic data to assess the stability of the one or more injectors.

(12) In another aspect, a system for subscale testing of rocket injector stability includes a chamber, a throat, and an injector plate. The chamber has a sidewall extending axially. The throat is at a downstream end of the chamber. The injector plate is removably positioned at an upstream end of the chamber. The injector plate includes one or more injector elements configured to inject one or more propellants into the chamber. The injector plate is one of a plurality of different injector plates configured to be removably attached to the chamber. The injector plate is continuously moveable in an axial direction to continuously vary a combustion volume of the chamber located between the injector plate and the throat.

(13) Various embodiments of the various aspects may be implemented. In some embodiments, a positioning of the throat remains fixed. In some embodiments, the system includes an actuator configured to axially move the injector plate. In some embodiments, the throat has an inner diameter that decreases from a first inner diameter to a second inner diameter in an upstream direction and increases from the second inner diameter to a third inner diameter in the upstream direction. In some embodiments, the system includes a sensor configured to detect pressure within the combustion chamber. In some embodiments, the throat defines a converging-diverging nozzle shape.

(14) In another aspect, a subscale rocket injector stability system includes a chamber, an injector plate, and a telescoping throat. The injector plate is removably attached to an upstream end of the chamber. The injector plate includes one or more injector elements configured to inject one or more propellants into the chamber. The injector plate is one of a plurality of different injector plates configured to be removably attached to the chamber. The telescoping throat is positioned at a downstream end of the chamber. The telescoping throat is continuously moveable in an axial direction to continuously vary a combustion volume of the chamber located between the injector plate and an upstream end of the telescoping throat.

(15) Various embodiments of the various aspects may be implemented. In some embodiments, in some embodiments the telescoping throat is positioned within the chamber. In some embodiments, a sidewall of the telescoping throat is positioned external to the chamber. In some embodiments, the system includes a first actuator configured to axially move the telescoping throat. In some embodiments, the system includes a second actuator configured to axially move the telescoping throat. The first and second actuator moving the telescoping throat simultaneously. In some embodiments, the telescoping throat includes a cooling channel. In some embodiments, the telescoping throat defines a converging-diverging nozzle shape. In some embodiments, in a downstream direction the telescoping throat has a first inner diameter that decreases to a second inner diameter and the second inner diameter increases to a third diameter. In some embodiments, the system includes a sensor configured to detect a pressure within the combustion volume.

(16) In another aspect, a method of subscale testing rocket injector stability includes injecting into a combustion chamber one or more propellants from one or more injector elements supported by a modular injector plate to cause combustion, continuously varying a combustion volume within the combustion chamber, and detecting acoustic data from within the combustion volume as the

combustion volume is varied.

(17) Various embodiments of the various aspects may be implemented. In some embodiments, continuously varying the combustion volume comprises continuously moving an external telescoping throat positioned at a downstream end of the combustion chamber. In some embodiments, continuously varying the combustion volume comprises continuously moving the modular injector plate. In some embodiments, continuously varying the combustion volume comprises continuously moving an internal telescoping throat through the combustion chamber. In some embodiments, the method includes replacing the modular injector plate with a different injector plate supporting one or more different injector elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only several embodiments in accordance with the disclosure and are not to be considered limiting of its scope, the disclosure will be described with additional specificity and detail through use of the accompanying drawings. In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the drawing, may be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated and make part of this disclosure.

(2) FIG. 1A is a perspective view of an embodiment of a subscale system for testing the stability of rocket injectors having a modular injector plate and continuously variable volume combustion chamber.

(3) FIG. 1B is a cross-sectional side view of an embodiment of a subscale system for testing the stability of rocket injectors having a modular injector plate and continuously variable volume combustion chamber with an axially moveable rounded piston.

(4) FIG. 2 is a cross-sectional side view of the piston of FIG. 1B having a rounded upstream end and a cooling channel.

(5) FIG. 2A is a cross-sectional view of an embodiment of a subscale system for testing the stability of rocket injectors having a linear actuator positioned below a combustion chamber.

(6) FIG. 2B is a cross-sectional view of an embodiment of a subscale system for testing the stability of rocket injectors having dual actuators in a first position.

(7) FIG. 2C is a cross-sectional view of the subscale system for testing the stability of rocket injectors of FIG. 2B with the dual actuators in a second position.

(8) FIG. 2D is a cross-sectional view of an embodiment of a subscale system for testing the stability of rocket injectors having a deflector and an actuator positioned downstream of the combustion chamber behind the deflector.

(9) FIG. 3 is a perspective view of a shell that is coupled to an upstream end of the piston of FIG. 1B.

(10) FIG. 4 is a cross-sectional side view of another embodiment of a subscale system for testing the stability of rocket injectors, having a modular injector plate and continuously variable volume combustion chamber, with an axially moveable piston with a planar upstream end.

(11) FIG. 4A is a side view of an axially moveable piston having an irregularly shaped upstream

end.

(12) FIG. 4B is a front view of the axially moveable piston of FIG. 4A.

(13) FIG. 4C is a side view of an axially moveable piston having a donut shaped upstream end.

(14) FIG. 4D is a front view of the axially moveable piston of FIG. 4C.

(15) FIG. 5A is a cross-sectional view of another embodiment of a subscale system for testing the stability of rocket injectors having an internal telescoping throat.

(16) FIG. 5B is a cross-sectional view of the subscale system of FIG. 5A illustrating an example embodiment of cooling channels within the telescoping throat.

(17) FIG. 5C is a cross-sectional schematic view of another embodiment of a subscale system for testing the stability of rocket injectors having an external telescoping throat.

(18) FIGS. 5D and 5E are cross-sectional views of another embodiment of a subscale system for testing the stability of rocket injectors having a telescoping injector plate.

(19) FIG. 6 is a flow chart showing an embodiment of a method for subscale testing of rocket injector stability.

(20) FIG. 7A is a data chart showing expected acoustic frequency data using a static subscale testing system.

(21) FIG. 7B is a data chart showing expected acoustic frequency data using a subscale testing system with a continuously variable combustion volume according to the present disclosure.

(22) FIG. 7C is a data chart showing example expected resonant frequency depending on combustion chamber length using a subscale testing system with a continuously variable combustion volume according to the present disclosure.

DETAILED DESCRIPTION

(23) The following detailed description is directed to certain specific embodiments for devices, systems, and methods related to testing the stability of subscale rocket injectors. In this description, reference is made to the drawings wherein like parts or steps may be designated with like numerals throughout for clarity. Reference in this specification to “one embodiment,” “an embodiment,” or “in some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrases “one embodiment,” “an embodiment,” or “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments.

Moreover, various features are described which may be exhibited by some embodiments and not by others. Similarly, various requirements are described which may be requirements for some embodiments but may not be requirements for other embodiments. Reference will now be made in detail to embodiments of the invention, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(24) A typical subscale testing system will have different acoustic modes than a full scale engine. This can be due to the flame and/or injector response not being coupled with acoustic modes across a range of frequencies. The subscale stability test systems and methods according to the present disclosure allow for understanding the acoustic response across a continuous range of frequencies using a continuously variable chamber volume, as further described.

(25) FIG. 1A is a perspective view of an embodiment of a subscale system **100** for testing the stability of rocket injectors. FIG. 1B is a cross-sectional side view of a portion of the subscale system **100**. The system **100** and other embodiments herein may be used to easily and quickly replace and test different injectors and to expose the injectors to acoustic modes of varying frequency that may cover the range of acoustic modes expected in a full scale engine. This may be done by continuously varying the volume of a combustion chamber during testing. Such variation in volume may be achieved by continuously moving a piston within the chamber, by continuously moving an injector plate within the chamber, continuously moving an internal telescoping throat

within the chamber, and/or continuously moving an external telescoping throat around the chamber. This provides for a number of benefits, including but not limited to, more efficient and quick testing, more accurate testing, and more cost-effective testing. Additionally, it eliminates the risk to other engine components that is present in full-scale engine testing.

(26) The system **100** may thus provide a combustion rig facility with a dynamically tunable downstream combustor and a modular injector interface. The modular interface allows multiple injector concepts to be quickly tested and assessed. The dynamically tunable downstream boundary enables the injector to be exposed to acoustic modes of varying frequency, covering the range of acoustic modes expected in a full scale engine. This allows the injector's stability characteristics to be screened without requiring a full scale engine test.

(27) The systems described herein thus overcome drawbacks of prior solutions to injector stability testing, such as traditional subscale test rigs producing different acoustic modes than a full scale engine, which may be due to the flame and/or injector response not being coupled with acoustic modes across a range of frequencies. Further, combustion stability is poorly understood, and lacks accurate, predictive analytical models. Due to stability being a system level phenomenon (i.e., a function of full-scale engine combustion chamber modes), stability issues often only become apparent during full-scale tests, where iteration on designs is costly and time-consuming, along with placing risk on other engine components. Traditional subscale test rigs are not considered useful to assess combustion stability due to fixed chamber dimensions that often correspond to acoustic modes which differ from all or some full scale combustor modes. Previous methods that have been studied introduce a driving signal of varying frequency using a siren. Another traditional alternative is to defer all stability testing to full-scale testing, with the inherent risk to cost and schedule that entails. All of these previous approaches do not sufficiently serve the industrial need to quickly and accurately screen and down-select injector concepts.

(28) Traditional approaches do not utilize common hardware and expose injectors to acoustic modes. In contrast, a longitudinal approach is the only subscale approach that supports a common set of combustor hardware regardless of the propellants, injector type, or operating conditions and subjects the injector to acoustic modes.

(29) The continuously tunable nature of the systems according to the present disclosure overcome these and other limitations. The system according to the present disclosure exposes the injector to all possible frequencies within the tunable range of the rig, thus effectively characterizing the stability response of the injector. The systems described herein further offer an effective component-level solution for characterizing stability and screening injector designs without requiring or risking other engine components, resulting in reduced cost and schedule for engine development programs. The systems described herein also reduce cost and risk for any new engine programs and allow design cycle times to be reduced. By rapidly screening multiple injector concepts at the component level, design iterations can happen much faster with the systems described herein and are decoupled from waiting for the readiness of other full-scale components. More technically ambitious designs can thus be pursued. The final full-scale engine test campaign can be approached having considerably reduced the risk of stability uncertainty.

(30) As shown in FIG. 1B, the system **100** may have a combustion chamber **104**. The system **100** may include an injector plate **108** removably attached at an upstream end of the combustion chamber **104**. The system **100** may include a moveable piston **112** located downstream from the injector plate **108**. As shown in FIG. 1A, “upstream” refers to a direction generally to the right and into the page of the figure as oriented, and “downstream” refers to a direction of movement of exhaust which is generally to the left and out of the page as oriented. As shown in FIG. 1B, “upstream” refers to a direction generally to the left in the figure as oriented, and “downstream” refers to a direction of movement of exhaust which is generally to the right as oriented.

(31) The injector plate **108** may be a structural holder configured to carry a single or plurality of injector elements **110**. The injector plate **108** may be metallic. The injector plate **108** may be

generally cylindrically shaped. In some embodiments, the injector plate **108** may be modular. For example, the injector plate **108** may be one among a plurality of different injector plates that may be attached to the system **100**. In some embodiments, the injector plate **108** may be attached via a manifold **109**. The manifold **109** may be a structural component configured to attach with the injector plate **108**. The manifold **109** may have an annular shape with an opening therethrough configured to partially receive the injector plate **108** therein. The injector plate **108** may have a reduced width, downstream portion received within the manifold **109**. In some embodiments, the system **100** may not include the manifold **109**.

(32) The injector elements **110** may be configured to inject one or more propellants, such as a fuel and an oxidizer, into the combustion chamber **104**. In embodiments where the injector plate **108** has a single injector element **110**, a second injector element **110** may be located elsewhere in the system, such as adjacent to the injector plate **108**. Each injector element **110** may inject a fuel or an oxidizer. The injector plate **108** may comprise one or more first injector elements **110** having a first set of injection characteristics, and the injector plate **108** may be removed and replaced with a second, different injector plate comprising one or more second injector elements having a second, different set of injection characteristics relative to the first injector elements **110**. The first and second sets of injector elements **110** may have different lengths, widths, volumetric flow, flow channel configurations, etc. In some embodiments, the injector elements **110** may have differing arrangements and operations (e.g., shear coaxial, swirl, impinging, pintle). The injector elements **110** may have different mixture ratios of propellants. Each injector plate **108** may hold one, two, three, four, five, six, seven, eight, nine, ten, or more of the injector elements **110**.

(33) In some embodiments, the injector elements **110** may extend beyond the injector plate **108** and into the combustion chamber **104** (e.g., such as baffle elements). In some embodiments, the injector elements **110** may be recessed from the injector plate **108** such that the initial propellant interactions occur within the injector recess. In some embodiments, the injector plate **108** and/or the injector elements **110** may incorporate cavities to provide a damping function.

(34) The injector plate **108** may thus be interchangeable. Any number of different injector plates **108** may be configured to be attached to the combustion chamber **104** for testing. This allows for the testing of various injector plates **108** using a single testing system **100**. A first injector plate **108** may be tested and once testing is complete, the first injector plate **108** may be removed and the second injector plate may be attached, which may be tested and removed, and then a third injector plate may be attached, etc. In some embodiments, the various injector plates may have similar connecting structures for attaching to the same system **100**, for example to the manifold **109**. In some embodiments, the first injector plate **108** may have different internal attachments from a second injector plate, such that the two injector plates each receive different injectors. In some embodiments, the first injector plate **108** may have one or more first injector elements **110** removed and replaced by one or more second injector elements **110** that are different from the first injector elements and are installed into the same first injector plate **108**. Thus, the same injector plate **108** may be used for testing different injector elements **110**.

(35) In some embodiments, the system **100** may have an interchangeable cavity ring **111**. The ring **111** may form part of the combustion chamber **104**. The ring **111** may extend annularly around the chamber **104**. The ring **111** may be metallic. A radially inner surface of the ring **111** may be generally cylindrical. The ring **111** may be axially positioned between the injector plate **108** or manifold **109** and the piston **112**. “Axial” may be in directions parallel to a central axis as defined by a circularly shaped chamber **104**. The interchangeable ring **111** may be configured to provide a damping function. The interchangeable ring **111** may be configured to provide thermal protection or prevent melting or damage to the injector plate **108**. In some embodiments, the interchangeable ring **111** may be used to test film cooling and/or identify local injector extinction events. Rings **111** of different sizes may be used. For example, the thickness or material of the interchangeable ring **111** may be changed. In some embodiments, the system **100** may not include the ring **111**.

(36) In some embodiments, the ring **111** can be a simple annulus. The ring **111** can be configured to conduct flow to the downstream portion of the system **100**. In some embodiments, the ring **111** can include additional components. For example, the ring can include stability aids (e.g., acoustic cavities and bias flow elements), igniters, pulse-guns, and instrumentation such as thermocouples, high-speed pressure transducers, and optical measurements.

(37) The combustion chamber **104** may be cylindrical and have a sidewall **128** extending axially. The sidewall **128** may be metallic. The sidewall **128** may define a cylindrical inner surface and extend from an upstream end to a downstream end. An upstream end of the sidewall **128** may attach to or be adjacent the cavity ring **111** and/or the manifold **109**. The downstream end of the chamber **104** may be open, except for the piston therein, to allow exhaust to exit the combustion chamber **104** at the exit **119**.

(38) The combustion chamber **104** may have a diameter of about 3 inches, about 4 inches, about 5 inches, about 6 inches, about 7 inches, about 8 inches, or more or any values in between. The combustion chamber **104** may be at least partially formed by one or more plates **122**. Examples plates **122** removed from the system **100** are visible in FIG. 1A. The plates **122** may have a constant inner diameter to produce a constant inner diameter of the combustion chamber **104**. In some embodiments, the plates **122** may comprise varying diameters to form a combustion chamber with varying diameters. For example, a first plate **122** may have a first diameter and a second plate **122** may have a second diameter that is larger or smaller than the first diameter. Any combination of diameters may be used.

(39) The plates **122** may be structural segments defining an interior cylindrical opening. The plates **122** may be metal, composite, or other suitable materials. The sidewall **128** may be located along the inner surfaces of the plates **122**. The sidewall **128** may be formed by the inner surfaces of the plates **122**, or the sidewall **128** may be a separate structure attached to the plates **122**. There may be one, two, three, four or more plates **122**. The axial length of the combustion chamber **104** may be determined by the number of plates **122** and the width of the plates **122**. However, the axial length of a region **105** of combustion within the chamber **104** upstream of the piston may be varied, as further described.

(40) The system **100** may include one or more calorimeter bodies **123**. The plates **122** forming the combustion chamber **104** may comprise the calorimeter body **123**. The calorimeter body **123** may be configured to measure heat flux. For example, water may flow through the calorimeter body **123** to absorb the heat. The calorimeter body **123** may be used to assess the injector performance. The calorimeter body **123** may be within or form the sidewall **128**.

(41) The piston **112** may be at least partially located inside the combustion chamber **104**. The piston **112** may be inserted at a second downstream end opposite the first upstream end that is attached to the injector plate **108**. The piston **112** may be located at least partially upstream of the downstream end of the chamber **104**. The piston **112** may include a head **120** at an upstream end thereof. The head **120** may be located within the chamber **104** between the sidewall **128**. The head **120** may be located upstream of the downstream end of the chamber **104**. The head **120** may be moved axially during testing, as further described.

(42) The combustion chamber **104** may have a combustion volume or region **105** configured to have a variable volume. The region **105** may be located within the combustion chamber **104** upstream of the head **120** of the piston **112** or portions thereof. The region **105** may extend from an upstream end of the chamber **104**, such as at the injector plate **108** or manifold **109**, to an upstream end **121**, a middle portion **125**, and/or a downstream end **127** of the piston **112**. The volume of the region **105** may be changed by translating the piston **112** in an axial direction within the combustion chamber **104**. For example, with reference to the positioning of FIG. 1B, the region **105** of the combustion chamber **104** will reduce in volume when the piston **112** is moved upstream (to the left as oriented) and increase in volume when the piston **112** is moved downstream (to the right as oriented). The relative directions described herein as oriented in the figures are not

limiting. For example, in other embodiments, the orientation of the piston **112** and the injector plates **108** may be switched, where the piston **112** is on the left and the injector plate **108** is on the right.

(43) The system **100** may include an actuator **116**, shown schematically in FIGS. **1B** and **4**. The actuator **116** may be configured to move the piston **112** in axial directions (e.g., left and right as oriented) within the combustion chamber **104**. The movement of the piston **112** may continuously vary the volume of the region **105** of the combustion chamber **104**. The actuator **116** may be mounted radially to one side of the system **100** in order to avoid damage caused by the exhaust exiting the combustion chamber **104** at the exit **119**. The actuator **116** may be a hydraulic actuation system. The actuator **116** may continuously move the piston **112** during testing. By “continuous” it is meant the piston **112** is not limited to discrete locations within the chamber but may move along the chamber **104** for collecting mode data along a continuum. The piston **112** may be translated across a range of about 4 inches to about 60 inches. The range may be from 4 inches to 60 inches. The range may be determined based on screening an injector concept across a frequency range consistent with the natural acoustic modes of the rocket engine injector. The speed of the motion of the piston **112** may be a value which allows a quasi-steady assumption of the operating conditions of the combustion chamber **104** to remain valid. In some embodiments, the speed may be dependent upon the limitations of a test facility. Modes and injector responses may thus be analyzed for combustion volumes of the region **105** having a continuously changing size. In other words, the relationship between axial position of the piston **112** and volume of the region **105** may be generally smooth with no sharp jumps, and with a continuous first derivative across all volumes tested.

(44) In some instances, testing may include rapidly translating the piston **112** over a short distance. This type of testing could create an artificial disturbance, for example an acoustic impulse similar to that which is created from a bomb or pulse gun shot. The results of this testing may be used for determination of stability ratings.

(45) The piston **112** may have an extension rod **124**. The head **120** may attach to an upstream end of the rod **124**. The rod **124** may be axially elongated and extend between the head **120** and the actuator **116**. In some embodiments, the head **120** is replaceable with different shaped and sized heads. In some embodiments, the head **120** and rod **124** are a single, monolithic structure.

(46) The head **120** of the piston **112** may have various shapes. As shown in FIG. **1B**, the head **120** may be rounded, for example bulbous. In some embodiments, the head **120** may be flat or planar, for example as shown in FIG. **4**. The head **120** may have a variable width to provide a chokepoint within the combustion chamber **104** at an annular gap **126**. The annular gap **126** may be a portion of the flow path of the combustion exhaust in an annular, e.g., circular, region defined between the piston **112** and the sidewall **128**. The annular gap **126** may be a region of minimal cross-sectional area of the chamber **104**. The annular gap **126** may be a location of minimal cross-sectional area between the piston **112** and the sidewall **128**.

(47) A cross-sectional area of the combustion flow path within the chamber **104** may be constant in the downstream direction from the injector plate **108** to the upstream end **121** of the head **120** of the piston **112**. The cross-sectional area of the flow path may vary axially along the piston head **120**. The piston head **120** may have a variable width in the axial direction. In some embodiments, the piston **112** increases in cross-sectional area in the downstream direction from the upstream end **121** of the head **120**, to a maximum cross-sectional area at the middle portion **125** that partially defines the annular gap **126**. The piston may decrease in cross-sectional area in the downstream direction from the middle portion **125** or annular gap **126** to the downstream end **127**. The rate of increase of the cross-sectional area from the upstream end **121** to the middle portion **125** may be greater on an absolute basis as compared to the rate of decrease of the cross-sectional area from the middle portion **125** to the downstream end **127**. The axial length between the upstream end **121** and the middle portion **125** may be greater than the axial length from the middle portion **125** to the

downstream end **127** as shown.

(48) The size and shape of the annular gap **126** may be adjusted by using different sizes and shapes of the piston head **120**. The shape and size of the piston head **120** may be selected to match a desired engine contraction ratio. The engine contraction ratio may be the ratio of the combustion chamber cross sectional area, for example in the combustion region **105** upstream of the piston head **120**, to the minimum cross-sectional area of the flow path through the annular gap **126**. This region of minimum cross-sectional area may be at the annular gap **126** between the chamber sidewall **128** and the middle portion **125** of the piston head **120**.

(49) The cross-sectional area of the combustion flow path in the downstream direction may thus be constant, reduce to a minimum at a chokepoint of the annular gap **126**, and then increase. The piston **112** and the sidewall **128** may in some embodiments define a volume having an inverted nozzle shape. The annular gap **126** may act as the chokepoint, such that the system **100** mimics the behavior of a converging-diverging nozzle with a throat. The annular gap **126** may be a location where the flow velocity is Mach 1 or about Mach 1.

(50) The system **100** may include one or more sensors **113**. The sensors **113** may be high frequency pressure transducers. The sensors **113** may be mounted at or near the injector plate **108**. In some embodiments, the sensors **113** may be positioned at the axial midpoint of the ring **111**. The sensors **113** may be positioned circumferentially about the ring **111**. In one example non-limiting embodiment, three sensors **113** can be positioned circumferentially at the axial midpoint of the ring at 0 degrees, 135 degrees, and 225 degrees. In some embodiments, there may be a plurality of the sensors **113** distributed annularly about an inner circumference, and/or axially along the axial length of, the sidewall **128** of the combustion chamber **104**. In some embodiments, the sensors **113** may be distributed axially in the chamber **104**. For example, a sensor **113** may be positioned at an axial midpoint of one or more of the plates **122**, as shown in FIG. 4. The sensor **113** may be positioned at the 0 degree position. The sensors **113** may be carried by the ring **111**. In some embodiments, the sensors **113** may be positioned on the piston **112**. The acoustics resulting from the combustion may bounce back from the head **120** of the piston **112**. For example, the head **120** of the piston **112** may act as an acoustic end of the combustion chamber **104**. The soundwaves or pressure waves from combustion may reflect from the head **120** of the piston **112**. The one or more sensors **113** may detect such acoustic or pressure waves.

(51) FIG. 2 is a close-up, cross-sectional view of the head **120** of the piston **112** of FIG. 1B having a shell **132**. FIG. 3 is a perspective view of the shell **132** in isolation. The shell **132** may be a structural component attached to the upstream end **121** of the piston **112**. In some embodiments, the shell **132** and the head **120** may be a monolithic structure. For example, the shell **132** and the head **120** may be additively manufactured as a single, continuous structure. The shell **132** may extend along the upstream end **121**, in the downstream direction, and terminate at the middle portion **125** as shown. In some embodiments, the shell **132** may terminate prior to or after the middle portion **125** in the downstream direction. The shell **132** may be a thin-walled structure.

(52) The piston **112** may include a one or more cooling channels **130**. The shell **132** may partially define the cooling channels **130**. Part of the cooling channels **130** may be formed at the head **120** of the piston **112** by the shell **132**, as shown in FIG. 2. The shell **132** may be coupled to the head **120**. The head **120** may have a receiving portion **134** which may include an indentation **139** configured to receive a protrusion **136** of the shell **132**. The protrusion **136** may extend annularly and in the downstream direction, with the receiving portion **134** defining the indentation **139** that correspondingly extends annularly and in the downstream direction. Coolant may flow through the channels **130** in the direction indicated by the arrows **137**.

(53) The cooling channel **130** may extend along a central longitudinal path of the head **120** of the piston **112**. The cooling channel **130** may extend from the extension rod **124** to the path within the head **120**, and into spaces between the head **120** and the shell **132** and/or within the shell **132**, as shown by the arrows **137** in FIG. 2. The cooling channel **130** may be configured to carry a coolant.

The coolant may travel through the cooling channel **130** starting in the extension rod **124** and then out the cooling channel **130** between the shell **132** and the head **120** of the piston **112**. In some embodiments, the coolant may flow back in the opposite direction of the arrows, e.g. through return channels. In some embodiments, the coolant may exit the head **120**, e.g. flow into open air or into additional tubing attached thereto. In some embodiments, the coolant may have a return flow in a direction back through the extension rod **124** (e.g., to the right as oriented in the figure). The head **120** of the piston **112** may define an acoustic boundary and a known contraction ratio. The cooling channel **130** may assist in preventing damage to the piston **112** during use. For example, the cooling channel **130** may prevent or reduce wear and tear on the piston **112**, allow for more accurate data to be collected, and/or prevent or reduce the chance of the piston **112** melting due to the heat exposure. Additionally, the cooling channel **130** may be advantageous in that it allows for longer periods of testing with the piston **112** translated during hot fire, which in turn may improve the resolution of frequency screening.

(54) As shown in FIG. 3, the shell **132** may have a plurality of grooves or channels **138** (for clarity, only some of the channels **138** are labelled in FIG. 3). The channels **138** may be grooves on an inner surface of the shell **132**. The channels **138** may extend from the open downstream end of the shell **132** toward the protrusion **136** with a depth partially through the thickness of the shell **132**. The channels **138** may define terminal portions of the cooling channels **130**. There may be ten or more, twenty or more, thirty or more, or forty or more of the channels **138**. Flow channels may be defined partially by the channels **138** and the outside surface of the head **120** of the piston **112**.

(55) FIGS. 2A-2D depict various embodiments of the system **100** having various actuators. As shown in FIG. 2A, in some embodiments, the system **100** may include an actuator **116A**, which includes a single, external linear actuation mechanism. FIGS. 2B and 2C show the system **100** with an actuator **116B** which includes dual external linear actuation components. FIG. 2D shows the system **100** with an actuator **116C** positioned downstream and aligned with the combustion chamber **104**. Some of the features of the actuators **116A**, **116C** are similar to the features of actuator **116** in FIGS. 1 and 4, except as otherwise described. Therefore, the structure and description for the various features of the actuator **116** and how it is operated and controlled in FIGS. 1 and 4 are understood to apply to the corresponding features of the actuator **116A** in FIG. 2A and **116C** in FIG. 2D, except as described below. Though the features below are described in connection with the actuator **116A**, the features also relate as applicable to other actuators, such as actuators **116**, **116B**, **116C** disclosed herein.

(56) As shown in FIG. 2A, the actuator **116A** may include a single linear actuation component positioned to one side of the system **100**, such as below the combustion chamber **104**. In some embodiments, the actuator **116A** may be positioned adjacent an outer wall of the combustion chamber **104** and upstream of the exhaust exit opening. The positioning of the actuator **116A** below or adjacent an outer wall of the combustion chamber **104** may be advantageous in that it protects the actuator **116A** from exhaust exiting the system **100** at the exit **119**. In some embodiments, the actuator **116A** may be positioned a distance DI from an outside surface **106** of the combustion chamber **104**. In some embodiments, the actuator **116A** may be coupled to the outside surface **106** of the combustion chamber **104**.

(57) The actuator **116A** may include an extension rod **115** that is configured to continuously move the piston **112** within the chamber **104**. The extension rod **115** may be coupled directly to the extension rod **124** of the piston **112**. The extension rod **115** may be coupled to a lateral portion **117** of the piston **112** that extends laterally or radially away from a central axis defined by the chamber **104**. The lateral portion **117** may extend generally perpendicular to the extension rod **124** of the piston **112**. In some embodiments, the lateral portion **117** and the extension rod **124** of the piston **112** may be a single integral or unitary part. During testing, the extension rod **115** of the actuator **116A** may push and pull the lateral portion **117** of the piston **112** to continuously change the positioning of the piston **112** within the chamber **104**. The movement of the piston **112** may

continuously vary the volume of the region **105** of the combustion chamber **104** according to the present disclosure.

(58) FIGS. **2B** and **2C** show an embodiment of the system **100** that includes dual actuators **116B**. Some of the features of the actuators **116B** may be similar to the features of actuator **116** in FIGS. **1** and **4** and the actuator **116A** in FIG. **2A**. Therefore, the structure and description for the various features of the actuators **116** and **116A** and how they are operated and controlled in FIGS. **1** and **4**, and FIG. **2A**, respectively, are understood to apply to the corresponding features of the actuators **116B** in FIGS. **2B** and **2C**, except as described herein. Though features may be described in connection with the actuators **116B**, the features may also apply to other actuators, such as the actuators **116**, **116A** disclosed herein.

(59) As shown in FIGS. **2B** and **2C**, the system **100** may include two actuators **116B** (i.e., a dual actuator system). FIG. **2B** illustrates the piston **112** in a first linearly contracted position, while FIG. **2B** illustrates the piston **112** in a second linearly extended position that is different than the first. The dual actuator system may be advantageous in that the use of two actuators **116B** provide additional structural stability and alignment control. One or both of the actuators **116B** may include the same features as the actuator **116A** described herein. The actuators **116B** may be mounted to an outside surface **106** of the combustion chamber **104**. A first actuator **116B** may be positioned directly opposite, e.g. rotationally 180 degrees from, a second actuator **116B**. One actuator **116B** may be positioned on top of the combustion chamber **104** and the second actuator **116B** may be positioned on the bottom of the combustion chamber **104**. The actuators **116B** may be positioned on opposite sides of the combustion chamber **104**. The actuators **116B** may be driven by a single motor. The use of a single motor may be advantageous in that it allows for the operation of the dual actuators **116B** to be synchronized. In some embodiments two motors may be used. In some embodiments, one of the actuators **116B** may actively move the piston **112** and the other of the actuators **116B** may be passive and provide support to the linearly moving lateral portion **117**.

(60) The actuators **116B** may each include the extension rod **115**, as similarly described with reference to FIG. **2A** above. The extension rods **115** may be coupled to or form a single integral or unitary part with the lateral portion or portions **117** of the piston **112**. As shown in FIG. **2A**, in some embodiments, cooling channels (e.g., cooling channels **130**) may follow a path represented by the arrows and extend through the lateral portion **117** and to/through the piston **112** and the extension rod **124** of the piston **112**. As shown in FIGS. **2B** and **2C**, in some embodiments, the cooling channels may follow a path represented by the arrows through the extension rod **115**, lateral portion **117**, the extension rod **124**, and the head **120** of the piston **112**. The cooling channels may extend from each end of the extending portion **117** and meet where the extension rod **124** extends from the lateral portion **117**. The cooling channel may travel down the extension rod **124** to the head **120** of the piston **112**. The cooling channel may have one or more exits that exit at the head **120** of the piston **112**, as shown in FIGS. **2B** and **2C**. The cooling channel may have a return channel with one or more exits in other locations, such as at the top of the lateral portion **117**, as shown in FIG. **2A**. The coolant may exit into additional tubing attached thereto. The size of the actuators **116B** may be determined, at least in part by, the size of the chamber **104**, the size and/or path of cooling channels, and/or any instrumentation access that is required.

(61) The cross-sectional area of the chamber downstream of the piston head **120** may be constant. In some embodiments, the cross-sectional area may vary. In some embodiments, there may be one or more fins **118** located downstream of the piston head **120**, for example as shown in FIGS. **2B** and **2C**. The one or more fins **118** may extend radially outward from the piston **112**. The one or more fins **118** may be wider at a base portion **190** (as labeled in FIG. **2C**) located near the piston **112** and narrower at a tip portion **191** (as labeled in FIG. **2C**) located away from the piston **112**. The one or more fins **118** may taper, e.g. decrease in width, in the outward radial direction. The one or more fins **118** may have an angled surface on an upstream facing side of the one or more fins **118** and/or the one or more fins **118** may have an angled surface on a downstream facing side of the one

or more fins **118**. There may be a single, continuous fin **118** extending circumferentially and continuously around the piston **112**. In some embodiments, the one or more fins **118** may protrude radially outward from the piston **112** no more than 25%, no more than 50%, or no more than 75%, of the distance between the piston **112** and an opposing interior surface of the sidewall **128**. The size of the one or more fins **118** may be application dependent. The one or more fins **118** may assist in maintaining the alignment of the piston head **120** within the combustion chamber **104**, for example along a longitudinal axis of the combustion chamber **104** or along a parallel axis thereof. The one or more fins **118** may assist in providing stabilization for the extension rod **124** as the extension rod **124** translates within the combustion chamber **104**. In some embodiments, the one or more fins **118** may be water cooled (e.g., comprise cooling channels). The one or more fins **118** may be a metallic material. In some embodiments, the one or more fins **118** may be a high temperature material. For example, the one or more fins **118** may be a carbon graphite.

(62) FIG. 2D shows an embodiment of the system **100** that includes the actuator **116C** positioned downstream of the combustion chamber **104**. The actuator **116C** may have the same or similar features as the actuator **116** in FIGS. 1 and 4, the actuator **116A** in FIG. 2A, and/or the actuator **116B** in FIGS. 2B and 2C, and vice versa. Therefore, the structure and description for the various features of the actuators **116**, **116A**, and **116B** and how they are operated and controlled in FIGS. 1 and 4, FIG. 2A, and FIGS. 2B and 2C, respectively, may apply to the corresponding features of the actuator **116C** in FIG. 2D, except as otherwise described herein.

(63) As shown in FIG. 2D, the system may include the actuator **116C** positioned downstream of the exit **119** of the combustion chamber **104**. In some embodiments, the actuator **116C** may be axially aligned with the exit **119** in a region of exhaust emitted from the combustion chamber **104**. The actuator **116C** may be protected by a deflector **150**. The deflector may be a shield extending circumferentially about the extension rod **124**. The deflector **150** may be positioned such that the exhaust exiting the combustion chamber **104** is deflected away from the actuator **116C**. The deflector **150** may prevent or reduce the risk that exhaust exiting the combustion chamber **104** damages the actuator **116C**. In some embodiments, an upstream facing side of the deflector **150** may be positioned at an angle. For example, the deflector **150** may extend outward and away from the extension rod **124**. The angled positioning of the deflector **150** may control the direction in which the exhaust exiting the combustion chamber **104** is deflected. In some embodiments, the extension rod **124** extends through an opening **151** of the deflector **150**. The opening **151** may extend through the thickness of the deflector **150**. The opening **151** may be sized such that a gap between the deflector **150** and the extension rod **124** is minimized to reduce and/or prevent exhaust from reaching the actuator **116C**. The opening **151** may be sized to allow the extension rod **124** to be translated through the deflector **150**, while also preventing or reducing the amount of exhaust traveling therethrough. In some embodiments, the deflector **150** may include cooling channels configured to cool the deflector **150**.

(64) FIG. 4 is a perspective cross-sectional view of an embodiment of another system **101** for subscale testing of rocket engine injectors. The system **101** may be the same as the system **100** except for the piston **114**. The piston **114** may have a head **131** and the extension rod **124**. The head **131** may have a different shape and other features as compared to the head **120**, as further described.

(65) As shown in FIG. 4, the most upstream portion of the head **131** may be flat or planar. The head **131** may have a rounded, e.g., circular, diameter. The diameter of the head **131** may be smaller than the diameter of the combustion chamber **104**. The diameter of the extension rod **124** may be smaller than the diameter of the head **120**. The head **131** may have a thickness T in the axial direction. The thickness T may extend from the most upstream portion of the head **131** to the start of a transition to the extension rod **124**. In some embodiments, the transition may be a gradual curve, with a varying diameter. In some embodiments, the transition may form a right angle, wherein a surface of the head **131** is perpendicular to the extension rod **124**. The diameter of the

head **131** may form the annular gap **126** between the outer diameter of the head **131** and the inner diameter of the combustion chamber **104**.

(66) The piston **114** may have the cooling channel **130**. The cooling channel **130** may extend through the extension rod **124** and into an interior of the head **131**. The cooling channel **130** can extend around a solid central portion **129** of the extension rod **124**. The solid central portion **129** can extend from the head **131** through the extension rod **124**. The cooling channel **130** may be configured to carry a coolant. The coolant may travel through the cooling channel **130** starting in the extension rod and to the head **131**.

(67) FIGS. **4A** and **4B** are side and front views respectively of an alternative embodiment of the piston (identified as a piston **301**). Embodiments of the piston **301** may include any of the features discussed above or below with reference to other pistons (e.g., piston **112**), and should not be limited to the particular embodiments discussed. For example, features of one embodiment may be combined with features of another embodiment. The particular modifications shown in FIGS. **4A** and **4B** will now be discussed in detail, and features not discussed will be understood to be similar, or identical, to those discussed above. Some or all of the modifications discussed below may be incorporated into the above disclosure.

(68) The head **120** of the piston **301** may have a triangular shape. The head **120** of the piston **301** and an inner wall **133** of the combustion chamber **104** may define one or more open areas or channels **360**. The open areas or channels **130** can define flow paths for the combustion exhaust. The head **120** may include a plurality of radially extending segments **367**, for example three segments **367**. The segments **367** may have outer circumferential walls **361**, which may be configured to contact the inner wall **133** of the combustion chamber **104**. Exhaust may be prevented from flowing around the exterior of the piston **301** at the interfaces of the walls **361** and the inner wall **133** of the chamber **104**. All exhaust may flow through the channels **360**. The plurality of walls **361** may have a curvature that corresponds to the curvature of the inner wall **133** of the combustion chamber **104**. The plurality of walls **361** may be circumferentially spaced around the head **120** of the piston **301**. The segments **367** may include radially outward facing inner walls **362**. Each adjacent wall of the plurality of walls **361** may be connected by the curved wall **362**. The curved walls **362** may define a portion of one of the open areas **360**. The plurality of walls **361** and the curved walls **362** may all taper to a blunt end **363** at the upstream most end of the head **120** of the piston **301** and taper toward the extension rod **124** at the end opposite the upstream end, for example as shown in FIG. **4A**.

(69) FIG. **4C** is a side view of an alternative embodiment of the piston (identified as a piston **302**) and FIG. **4D** is a front view. Embodiments of the piston **302** may include any of the features discussed above or below with reference to other pistons (e.g., piston **112**, piston **301**), and should not be limited to the particular embodiments discussed. For example, features of one embodiment may be combined with features of another embodiment. The particular modifications shown in FIGS. **4C** and **4D** will now be discussed in detail, and features not discussed will be understood to be similar, or identical, to those discussed above. Some or all of the modifications discussed below may be incorporated into the above disclosure.

(70) The head **120** of the piston **302** may have a donut shape cross-section that is elongated axially. The head **120** of the piston **302** may include a circumferential wall **365** that defines an internal channel **364** that extends through the interior of the head **120** of the piston **302** in the same general direction that the combustion chamber **104** extends, e.g. axially. The wall **365** may form a generally cylindrical shape. The wall **365** may have a thickness **T1** that increases from a first upstream end of the head **120** of the piston **302** to a generally central location along the length of the head **120** of the piston **302** and then decreases toward a second opposite, downstream end of the head **120** of the piston **302**. An upstream, transverse width **W1** of the head **120** at or near the upstream end of the head **120** may be smaller than a central width **W2** of the head **120** located at or near the central region of the head **120**, which may be larger than a downstream width **W3** of the head **120** at or

near the downstream end of the head **120**. In some embodiments, the transverse width of the head **120** may continuously increase in the downstream direction from the first upstream end of the head **120** to the generally central location of the head **120** (for example, from W1 to W2) and then continuously decrease toward the downstream second end of the head **120** (for example, from W2 to W3).

(71) The head **120** of the piston **302** may have an outer diameter that is less than the inner diameter of the combustion chamber **104**. There may be an annular gap **126** defined between the wall **365** of the head **120** and the inner surface **135** of the wall **128** of the chamber **104**. A radial dimension of the annular gap **126** may be a minimum at the central location of the head **120**, for example at the width W2. In some embodiments, the wall **365** of the head **120** may contact the inner surface **135** of the wall **128** of the chamber **104** to prevent exhaust from flowing around the exterior of the head **120**. Some or all exhaust may flow through an interior channel **364** defined by the head **120**. The interior channel **364** may be an opening extending through the head **120**. The piston **302** may be coupled to one or more actuators **116**. The one or more actuators **116** may be coupled to the wall **365**.

(72) The interior channel **364** of the head **120** may have a variable width in the axial direction. The interior channel **364** may provide a chokepoint within the head **120** at a throat **366**. The throat **366** may be a portion of the interior channel **364** within the head **120** that has a minimal cross-sectional area. The throat **366** may be axially located at a central location of the head **120**. The throat **366** may be axially located upstream or downstream of a central location of the head **120**.

(73) FIGS. 5A and 5B are cross-sectional views of an alternative embodiment of a subscale system **305** for testing the stability of rocket injectors that have internally telescoping features.

Embodiments of the system **305** may include any of the features of the testing systems discussed above or below, and should not be limited to the particular embodiments described. For example, features of one embodiment may be combined with features of another embodiment. The particular modifications shown in FIGS. 5A and 5B will now be discussed in detail, and features not discussed will be understood to be similar, or identical, to those discussed elsewhere herein. Some or all of the modifications discussed with respect to FIGS. 5A and 5B may be incorporated into the other embodiments described herein.

(74) The system **305** may have an internal telescoping throat **300**. The telescoping throat **300** may function and be translated similarly to the piston **112** described herein. While the telescoping throat **300** is coupled to the system **305** via dual actuators **116B**, any of the actuators described herein may be used. The telescoping throat **300** may have an outer diameter at an upstream end that is sized to fit within the inner diameter of the combustion chamber **104**. The outer diameter of the telescoping throat **300** may be sized to slide within the combustion chamber **104** when the telescoping throat **300** is moved by the actuators **116B**. The movement of the telescoping throat **300** within the combustion chamber **104** may continuously vary the volume of the region **105** of the combustion chamber **104**.

(75) FIG. 5B is a closeup cross-sectional sideview of the telescoping throat **300** of the system **305**. The telescoping throat **300** may have a sidewall **304** that extends annularly around, and defines an inner surface, of the telescoping throat **300**. The sidewall **304** may define an inner width, e.g. inner diameter, of the telescoping throat **300** that decreases from an upstream end **308** of the telescoping throat **300** in the downstream direction to a minimum, and then increases in the downstream direction towards a downstream end **312** of the telescoping throat **300**. The sidewall **304** may shape an inner area **316** of the telescoping throat **300** such that it mimics the appearance of a converging-diverging nozzle.

(76) In some embodiments, the telescoping throat **300** may include cooling channels **130**. The paths of coolant flowing through the cooling channels **130** are illustrated by the arrows in FIG. 5B that extend through the cooling channels **130**. Coolant may enter the cooling channels **130** through one or more openings **320**. The coolant may then travel toward the upstream end **308** of the telescoping

throat **300** through a portion of the cooling channel **130** that is defined between the inner diameter of the combustion chamber **104** and the outer diameter of the telescoping throat **300**, where it may reach a coolant seal **324**. The coolant seal **324** may prevent or limit any coolant from flowing into the combustion chamber **104**. The coolant may then travel through the sidewall **304** and further toward the downstream end **312** of the telescoping throat **300**.

(77) The telescoping throat **300** may be advantageous in that it may provide lateral stability. The telescoping throat **300** may eliminate or reduce the risk of lateral movement which may result with a cantilevered piston. The telescoping throat **300** may also simplify chamber construction by allowing for a constant area exhaust length. The telescoping throat **300** also reduces the required actuator length, and may allow for all calorimeter bodies (e.g., calorimeter bodies **123**, shown in FIG. **1B**) to be sized for chamber heat flux.

(78) FIG. **5C** is a cross-sectional schematic view of an alternative embodiment of a subscale system **310** for testing the stability of rocket injectors having an external telescoping throat **328**.

Embodiments of the system **310** may include any of the features of other testing systems discussed herein, and should not be limited to the particular embodiments described. For example, features of one embodiment may be combined with features of another embodiment. The particular modifications shown in FIG. **5C** will now be discussed in detail, and features not discussed will be understood to be similar, or identical, to those discussed herein. Some or all of the modifications discussed with respect to FIG. **5C** may be incorporated into other embodiments described herein.

(79) The system **310** may have the external telescoping throat **328** configured to translate linearly in axial directions. The telescoping throat **328** may function and be translated similarly to the piston **112** described herein. One or more of the actuators **116A** may linearly translate the extension rod **115** and transfer linear motion to the telescoping throat **328** via the lateral portion **117**. While the telescoping throat **328** is coupled to the system **310** via the single actuator **116A**, any of the actuators described herein may be used. The telescoping throat **328** may have an axially-extending, cylindrical sidewall **329** that defines an inner diameter that is sized to fit around an outer diameter of the correspondingly cylindrical combustion chamber **104**. The sidewall **329** of the telescoping throat **328** may be moved along the outer diameter of a fixed, annular wall **128** of the combustion chamber **104** by the actuator **116A**. The movement of the telescoping throat **328** along the wall **128** of the combustion chamber **104** may continuously vary the volume of the inner region **105** of the combustion chamber **104**.

(80) The telescoping throat **328** may have an inner protrusion **332** that extends annularly around an inner surface of the telescoping throat **328**. The inner protrusion **332** may have a height, measured radially-inwardly, that gradually increases from a location closer to an upstream end **336** of the telescoping throat **328** to a maximum height, and then decreases in height from the maximum height towards a downstream end **340** of the telescoping throat **328**. The inner diameter of the telescoping throat **328** may be a minimum where the radially-inward height of the inner protrusion **332** is at its maximum. The inner protrusion **332** may shape an inner area **344** of the telescoping throat **328** such that it mimics the appearance of a nozzle. The inner protrusion **332** is shown schematically, and may have a smooth inner surface that mimics the shape of a converging-diverging nozzle, such as the shape of the inner area **316** shown in FIG. **5B**.

(81) The system **310** may include one or more cooling channels that flow coolant along paths denoted by the arrows **137**, **140** as shown. The telescoping throat **328** may include a cooling channel that flows coolant along the arrows **137** in a path that extends from the downstream end **340** of the telescoping throat **328** to the upstream end **336** of the telescoping throat **328**. The coolant in the telescoping throat **328** may travel along the arrows **137** through the inner protrusion **332**. The coolant in the combustion chamber **104** may flow through a cooling channel along the arrows **140** through a wall **128** of the combustion chamber **104**. The coolant may flow along the entire length of the combustion chamber **104**. The coolant may flow along a partial length of the combustion chamber **104**. In some embodiments, coolant may flow along a first portion extending

in a first direction along the length of the combustion chamber **104** and along a second portion extending in a second direction opposite the first direction along the length of the combustion chamber **104**. Coolant may exit the system **310** from the sidewall **329** or the wall **128** as shown. The coolant exiting the system may flow into additional tubes or other components external to the system **310**. The cooling channel flowing through the wall **128** of the combustion chamber **104** may be physically separate from the cooling channel that flows coolant in the telescoping throat **328**.

(82) FIGS. 5D and 5E are cross-sectional views of another embodiment of a subscale system **315** for testing the stability of rocket injectors having a telescoping injector plate **348**. Embodiments of the system **315** may include any of the features discussed herein, and should not be limited to the particular embodiments discussed. For example, features of one embodiment may be combined with features of another embodiment. The particular modifications shown in FIGS. 5D and 5E will now be discussed in detail, and features not discussed will be understood to be similar, or identical, to those discussed elsewhere herein. Some or all of the modifications discussed with respect to FIGS. 5D and 5E may be incorporated into other embodiments described herein.

(83) The system **315** may have the telescoping injector plate **348** configured to translate linearly within the chamber **104**. The telescoping injector plate **348** may include any of the features of the injector plate **108**, for example as described with respect to FIGS. 1A and 1B. The telescoping injector plate **348** may be sized to fit within the combustion chamber **104**. For example, an outer wall **370** of the telescoping injector plate **348** may have an outer diameter that is smaller than an inner diameter of the sidewall **371** of the combustion chamber **104**. The telescoping injector plate **348** may be removably positioned within an upstream end **372** of the combustion chamber **104**. The location of the upstream end **372** may change as the injector plate **348** translates.

(84) The telescoping injector plate **348** may be coupled to an actuator **352**. The actuator **352** may be configured to linearly translate the telescoping injector plate **348** through the combustion chamber **104** during use. The movement of the telescoping injector plate **348** may continuously vary the volume of the combustion region **105** of the combustion chamber **104**. For example, FIG. 5D illustrates the telescoping injector plate **348** in a first upstream position, where a length **L1** is defined from a downstream end of the injector plate **348** to a throat **356**, and which defines a larger volume of the region **105** as compared to that of FIG. 5E, which shows the telescoping injector plate **348** in a second downstream position relative to the first position and where a length **L2** extending from the downstream end of the injector plate **348** to the throat **356** is less than **L1**. In some embodiments, **L1** may be from about 10 inches to about 50 inches, from about 20 inches to about 40 inches, from about 25 inches to about 35 inches, about 30 inches, or 28 inches. In some embodiments, **L2** may be from about 5 inches to about 25 inches, from about 10 inches to about 30 inches, from about 15 inches to about 25 inches, about 20 inches, or 18 inches.

(85) The system **315** may include the throat **356**. The throat **356** may be fixed at the end of the downstream end **373** of the combustion chamber **104** opposite the telescoping injector plate **348**. The throat **356** may not move linearly or laterally with respect to the combustion chamber **104**. The throat **356** may remain stationary while the telescoping injector plate **348** is translated, thereby varying the volume of the combustion region **105**. The throat **356** may provide a converging-diverging cross-sectional area. The throat **356** may have an inner diameter that continuously decreases in a downstream direction from a first upstream inner diameter to a second smaller and minimum inner diameter, and then continuously increases in the downstream direction from the second inner diameter to a third larger inner diameter at the exit. The second diameter may thus be less than both the first and third diameters. The inner contour of the throat **356** may mimic that of a nozzle shape as described in various embodiments herein.

(86) The use of the telescoping injector plate **348** may be advantageous in that any variables due to translating the throat are removed as the throat remains fixed, all calorimeters (e.g., calorimeter bodies **123**) are sized for chamber heat flux, and the actuator is naturally positioned away from the

exhaust.

(87) Continuously varying the volume of the region **105** of the combustion chamber **104** according to the embodiments of the present disclosure may provide many benefits and advantages. These include the ability to support a system having a traverse length of over 10 inches, the ability to maintain a consistent throat area during operation by maintaining the static and dynamic alignment to the chamber as well as having fixed throat dimensions, and providing a well-defined acoustic boundary.

(88) FIG. **6** is a flow chart showing an embodiment of a method **200** of subscale testing of rocket injector stability. The method **200** may be performed by the systems **100**, **101**, **305**, **310**, **315**, or variations thereof.

(89) The method **200** begins with step **250** where propellants, such as a fuel and an oxidizer, may be injected into a combustion chamber (e.g., combustion chamber **104**). The propellants may be continuously injected into the combustion chamber. The fuel and oxidizer may be injected into the chamber by one or more injector elements (e.g., injector elements **110**).

(90) The method **200** may then move to step **251** wherein the propellants may be ignited within the combustion chamber. The fuel and oxidizer may combust within the combustion chamber. In some embodiments, the ignition of the propellants may optionally be performed during the injection step (e.g., step **250**). The modular injector plate **108** may support the injector elements **110**, and be replaced with different injector elements, etc. as described herein, after completing the method **200** for subsequent testing. The plate **108** may be attached to the system, for example to the manifold **109**. A piston or injector plate may be in a first location within the combustion chamber during step **250**.

(91) The method **200** then moves to step **252** where the combustion chamber volume is continuously varied by axial movement of a moveable component such as the piston (e.g., piston **112**), the throat (e.g. throat **300** or **328**), or the injector plate (e.g. injector plate **328**). In some embodiments, the combustion chamber volume may be continuously varied by continuously varying the length of the combustion chamber and/or the region **105** of combustion. The component may be moved from its first location during combustion of the injected fuel in step **250**. As described herein, the movement of the component may continuously vary the volume of the combustion region within the chamber. Thus different acoustic modes may be produced in step **252** due to movement of the component. The component may be smoothly moved from a first axial location to a second downstream axial location, or vice versa. The component may be in multiple locations between the first and second locations during combustion.

(92) In some embodiments, step **252** may further include flowing a coolant through a cooling channel (e.g., cooling channel **130**) within the component. As described herein, the coolant can flow through a piston and exit through a gap between the shell (e.g., shell **132**) and the head **120** of the piston **112**. In other embodiments, coolant may flow through the various throats or combustion chambers, as described herein. The method **200** may also include replacing the injector plate (e.g., injector plate **108**) with a different injector plate after completion of the method **200** for subsequent testing. The method **200** may also include flowing exhaust from the combustion volume through an annular gap or flow path (e.g., annular flow path **126**) of the chamber. The combustion products may flow through various paths depending on the particular system utilized. For example, the combustion products may flow through the upstream, constant cross-sectional area portion of the chamber, into a decreasing cross-sectional area portion about an upstream portion of the piston, and then into an increasing cross-sectional area portion about a downstream portion of the piston, and then exit the chamber.

(93) The method then moves to step **254** where data may be acquired and/or detected. In some embodiments, the data acquired and/or detected may be acoustic data. For example, pressure data from within the combustion chamber may be detected as the component (piston, throat, or injector plate) is moved. One or more sensors (e.g., the sensors **113**) may be configured to detect the data at

one or more annular and/or axial locations within the chamber. In some embodiments, the data may be analyzed to assess the stability of the one or more injector elements. Various resonant frequencies may be identified as the component moves to characterize the injectors. Pressure may be detected over time. Changes in the pressure may be detected and analyzed. Acoustic modes may be identified based thereon to assess stability.

(94) FIGS. 7A and 7B are analytical data plots of instabilities and injector responses. Combustion chamber environments may create high levels of acoustic noise. The power spectrum of the acoustic pressure levels may show an oscillation, i.e., combustion instability, where sound pressure peaks with well-defined magnitudes are summed with the background noise. These peaks are correlated with the resonance frequencies of the combustion chamber, where the acoustic natural frequencies and the burning oscillations of the combustion chamber are coupled, which can cause the instabilities and consequent unexpected behavior such as efficiency loss or even explosion of the engine.

(95) FIG. 7A is a data chart showing expected acoustic mode data across various frequencies from a conventional “static” subscale testing system without a variable combustion chamber volume. Discrete peaks are shown for the subscale and full scale acoustic responses and instabilities. As shown, the sub-scale measured instability and acoustic response are at frequencies that are far different from the actual full scale engine instability and acoustic response. The static test system thus does not accurately predict the injector stability at full scale.

(96) In contrast, FIG. 7B is a data chart showing expected acoustic mode data using a subscale testing system according to the present disclosure with a continuously variable combustion volume (e.g., systems **100** or **101** or other embodiments described herein). FIG. 7B is representative of when the flame/injector response is coupled with acoustic modes across a range of frequencies as the downstream boundary is translated, spanning all of those anticipated at full-scale, for example, by moving a piston (e.g., piston **112**) within a combustion chamber (e.g., combustion chamber **104**), as described herein. The results as shown in FIG. 7B allows injectors (e.g., injector plate **108**) to be screened for stability issues in a consistent and efficient way well before full scale testing commences. As shown, the full scale measured instability envelope and acoustic modes can be accurately predicted based on the subscale testing results.

(97) FIG. 7C is a data chart showing expected resonant frequency depending on combustion chamber length using a subscale testing system with a continuously variable combustion volume according to the present disclosure. The data chart of FIG. 7C illustrates expected data in connection with the embodiment of the testing system **100** illustrated in FIG. 1B. As shown in FIG. 7C, it is expected that as the combustion chamber (e.g., combustion chamber **104**) increases in length, the resonant frequency will gradually decrease along a curve. The data chart of FIG. 7C illustrates various tangential modes for an example rocket engine and various longitudinal modes for an example subscale test system. In particular, the chart shows a first tangential mode **400**, a second tangential mode **401**, a third tangential mode **402**, a fourth tangential mode **403**, a fifth tangential mode **404**, and a sixth tangential mode **405** for the engine, and a first longitudinal mode **406**, a second longitudinal mode **407**, a third longitudinal mode **408**, a fourth longitudinal mode **409**, a fifth longitudinal mode **410**, and a sixth longitudinal mode **411** for the test system. The tangential mode may refer to the frequency of waves formed between two or more surfaces. The longitudinal mode may refer to the frequency of a particular standing wave pattern formed by waves in a confined area (e.g., the combustion chamber **104**).

(98) In addition to the benefits and advantages described herein, the system **100** may provide additional insight into injector response functions that is not available during full scale testing. For example, the controlled and variable boundary condition (variable chamber size) provides for more data. The system may also provide additional data on stability aids, such as the interchangeable cavity ring **111**. Further, the system **100** may allow for a quick response to stability issues that are identified during full scale testing as they could be quickly reproduced and tested using the system

(99) Various modifications to the implementations described in this disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the disclosure is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the claims, the principles and the novel features disclosed herein. The word “example” is used exclusively herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “example” is not necessarily to be construed as preferred or advantageous over other implementations, unless otherwise stated.

(100) Certain features that are described in this specification in the context of separate implementations also may be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also may be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

(101) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Additionally, other implementations are within the scope of the following claims. In some cases, the actions recited in the claims may be performed in a different order and still achieve desirable results.

(102) It will be understood by those within the art that, in general, terms used herein are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the

art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

Claims

1. A system for subscale testing of rocket injector stability, the system comprising: a chamber having a sidewall extending axially; an injector plate removably attached to an upstream end of the chamber, the injector plate structurally holding one or more injectors configured to inject one or more propellants into the chamber, the injector plate being one of a plurality of different injector plates configured to be removably attached to the chamber for interchanging the injector plate with a different injector plate of the plurality of different injector plates, the different injector plate structurally holding one or more different injectors; and a piston positioned at least partially within the chamber, the piston comprising a head on an upstream end thereof and an extension rod extending downstream from the head through a downstream exit of the chamber, wherein the piston is continuously moveable in an axial direction to continuously vary a combustion volume of the chamber located between the injector plate and the upstream end of the piston.
2. The system of claim 1, wherein the upstream end of the piston is rounded.
3. The system of claim 1, further comprising an actuator configured to axially move the piston.
4. The system of claim 1, wherein the piston further comprises a cooling channel.
5. The system of claim 4, further comprising a shell at the upstream end of the piston, wherein the cooling channel extends between the shell and the piston.
6. The system of claim 4, further comprising a shell at the upstream end of the piston, wherein the shell and the piston are monolithic.
7. The system of claim 1, wherein the piston and the sidewall of the chamber define a gap therebetween.
8. The system of claim 7, wherein the gap is an annular gap.
9. The system of claim 8, wherein the piston increases in cross-sectional area in a downstream direction from the upstream end of the piston to a maximum cross-sectional area at the annular gap.
10. The system of claim 9, wherein the piston decreases in cross-sectional area in the downstream direction from the annular gap.
11. The system of claim 1, further comprising a sensor configured to detect a pressure within the combustion volume.
12. The system of claim 1, further comprising a replaceable cavity ring forming part of the sidewall of the chamber.
13. A subscale rocket injector stability test system comprising: a chamber; a modular injector plate coupled to the chamber and comprising one or more injectors; and a piston positioned at least partially within the chamber and configured to move to vary a combustion volume of the chamber, the piston comprising a head on an upstream end thereof and an extension rod extending downstream from the head through a downstream exit of the chamber, wherein the piston and a sidewall of the chamber define an annular flow path around the piston configured to act as a chokepoint allowing exhaust to exit the chamber.
14. The system of claim 13, wherein the head on the upstream end of the piston is bulbous.
15. The system of claim 13, wherein the piston further comprises a cooling channel extending at least partially within the extension rod coupled to the piston and at least partially within the piston.
16. The system of claim 13, wherein a volume defined between the piston and the sidewall defines an inverted nozzle shape.
17. The system of claim 13, further comprising an actuator configured to axially move the piston to continuously vary the combustion volume.

18. A method of subscale testing rocket injector stability, the method comprising: injecting into a combustion chamber one or more propellants from one or more injectors supported by a modular injector plate to cause combustion; continuously moving a piston within the combustion chamber to continuously vary a combustion volume within the combustion chamber with an extension rod of the piston moving through a downstream exit of the combustion chamber; and detecting data from within the combustion volume as the piston moves.

19. The method of claim 18, further comprising flowing a coolant through a cooling channel within the piston.

20. The method of claim 18, further comprising replacing the modular injector plate with a different injector plate supporting one or more of different injectors.

21. The method of claim 18, further comprising flowing exhaust from the combustion volume through an annular gap located between the piston and a sidewall of the combustion chamber.

22. The method of claim 18, further comprising analyzing the data to assess the stability of the one or more injectors.
